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Simplifying complex content for students

Simplifying Complex Educational Content: Evidence-Based Strategies for Enhancing Student Comprehension

Quick Reference

Key Findings Table

Theme	Evidence-Based Strategies	Impact on Learning Outcomes	Citations
Constructivist Pedagogy	Scaffolding, inquiry-based learning, learning progressions, case studies	Improved comprehension, reduced misconceptions, higher engagement	1 2 3 4
Cognitive Load Theory (CLT)	Worked examples, segmentation, signaling, multimedia design principles	Reduced extraneous load, enhanced retention, better transfer of knowledge	5 6 7 8
Technology-Assisted Simplification	AI-driven adaptive learning, interactive multimedia, accessibility tools	Personalized learning, increased accessibility, improved engagement	9 10 11 12
Teacher Professional Development	Inquiry-based/scaffolded PD, collaborative design, reflective practice	Effective implementation, increased teacher confidence, sustained instructional change	13 14 15

Theme	Evidence-Based Strategies	Impact on Learning Outcomes	Citations
Gamification & Task Segmentation	Segmenting complex tasks, gamified learning, adaptive feedback	Enhanced motivation, improved skill acquisition, context-dependent cognitive load management	16 17 18
Accessibility & Inclusion	Assistive technologies, user-centered design, plain language	Improved outcomes for students with disabilities, increased content accessibility	19 20 21 22
Generative AI & Long-Term Outcomes	AI-generated adaptive content, blended learning approaches	Higher-order thinking, improved engagement, need for guided integration for knowledge retention	12 23 24

Direct Answer

Effective content simplification for student learning is best achieved by integrating constructivist pedagogical strategies (e.g., scaffolded learning, progressive complexity, inquiry-based activities) with cognitive load theory principles (such as worked examples, segmentation, and multimedia design). Technology, especially AI-driven adaptive platforms, further personalizes and breaks down complex content into more understandable, “human language” formats. For detailed evidence, see 3 7 9.

Study Scope

- **Time Period:** 2000–2024
- **Disciplines:** STEM, language education, health sciences, teacher education, educational technology
- **Methods:** Meta-analyses, experimental studies, systematic reviews, case studies, technology evaluations

Assumptions & Limitations

- Most evidence is drawn from formal K-16 education; transferability to informal or adult learning contexts may vary.
- Long-term impacts of AI-driven and generative tools are still under-researched.
- Teacher professional development quality and institutional support are critical but variable factors.
- Accessibility research is more robust for visual and cognitive disabilities than for other groups.

Suggested Further Research

- Longitudinal studies on the sustained impact of AI-driven adaptive content simplification.
- Optimization of teacher professional development for technology integration.
- Comparative studies of generative AI versus traditional instructional methods across diverse demographics.
- Research on content simplification for neurodiverse and multilingual learners.

1. Introduction

The Need for Content Simplification

Modern education faces the persistent challenge of making complex academic content accessible and engaging for all students. As curricula expand and knowledge domains become increasingly specialized, students often struggle with cognitive overload, disengagement, and persistent misconceptions. Simplifying content—translating "big" or abstract ideas into clear, relatable, and "human" language—is essential for:

- **Cognitive Accessibility:** Reducing working memory demands and supporting schema construction [1] [5].
- **Motivation:** Enhancing relevance and self-efficacy, especially for marginalized or diverse learners [2] [25].
- **Equity:** Ensuring all students, including those with disabilities or language barriers, can access and master core

concepts [19] [20].

Research demonstrates that effective content simplification is not about "dumbing down" material, but about strategic design, scaffolding, and personalization to foster deep understanding and lifelong learning [1] [3] [5].

2. Theoretical Frameworks

Constructivist Approaches and Active Learning

Constructivist pedagogy posits that learners build new knowledge by connecting it to prior experiences through active engagement, inquiry, and reflection. Key strategies include:

- **Scaffolding:** Providing temporary supports that are gradually removed as competence increases [3] [26].
- **Inquiry-Based Learning:** Encouraging students to explore, question, and investigate real-world problems [27] [28].
- **Learning Progressions:** Structuring content in developmentally appropriate sequences to address misconceptions and support conceptual change [29] [30].
- **Case Studies & Problem-Based Learning (PBL):** Using authentic scenarios to contextualize abstract concepts, especially effective in multicultural and multilingual classrooms [31] [32].

Synthesis: Constructivist strategies are empirically validated to simplify complex content, foster engagement, and reduce barriers for diverse learners [1] [2] [3] [4].

Applications of Cognitive Load Theory (CLT)

CLT provides a framework for designing instructional materials that align with human cognitive architecture. Its core principles include:

- **Worked Examples:** Step-by-step demonstrations that reduce extraneous load for novices [7] [33].
- **Segmentation:** Breaking content into manageable chunks to prevent overload [8] [34].
- **Signaling:** Highlighting essential information to direct attention [8] [35].
- **Redundancy & Split-Attention Effects:** Avoiding unnecessary repetition and integrating related information [6]

8 .

Synthesis: Applying CLT principles systematically enhances comprehension, retention, and transfer, especially in STEM and technical subjects 5 6 7 8 .

Technology-Assisted and AI-Driven Instruction

Emerging technologies offer scalable, personalized solutions for content simplification:

- **AI-Driven Adaptive Learning:** Platforms that tailor content, pacing, and feedback to individual learner profiles 9 10 .
- **Interactive Multimedia:** Animations, quizzes, and simulations that make abstract concepts concrete 36 37 .
- **Accessibility Tools:** Assistive technologies and user-centered design for students with disabilities 19 20 21 .
- **Generative AI:** Tools that autonomously adapt and generate educational content for diverse learning styles 12 23 .

Synthesis: Technology, when thoughtfully integrated, can break down complex content, personalize learning, and increase accessibility, though ethical and privacy considerations must be addressed 9 11 12 .

3. Methods & Data Transparency

This synthesis draws on:

- **Meta-Analyses:** Aggregating findings from hundreds of studies on constructivist pedagogy, CLT, and educational technology 1 5 25 .
- **Experimental & Quasi-Experimental Studies:** Evaluating interventions such as worked examples, scaffolding, and AI-driven tools in diverse educational settings 7 9 .
- **Systematic Reviews:** Assessing the effectiveness of teacher professional development, accessibility technologies, and gamification 38 39 .
- **Case Studies:** Documenting implementation in multilingual, multicultural, and inclusive classrooms 22 40 .

All claims are supported by direct citations to the underlying research base.

4. Critical Analysis of Findings

4.1 Pedagogical Strategies for Simplifying Content

Constructivist Approaches

- **Active Learning:** Engages students in constructing meaning, leading to deeper understanding and retention [1] [27].
- **Scaffolding:** Metacognitive and adaptive scaffolding are especially effective for diverse learners and those with learning disabilities [26] [41].
- **Learning Progressions:** Reduce misconceptions by aligning instruction with cognitive development and revisiting concepts with increasing complexity [29] [42].
- **Case Studies & PBL:** Lower academic barriers, foster collaboration, and improve comprehension in multicultural/multilingual settings [32] [40].

Teacher Professional Development

- **Inquiry-Based & Scaffolded PD:** Essential for effective implementation; collaborative design and reflective practice increase teacher confidence and instructional quality [38] [43].
- **Barriers:** Limited resources, time, and variable PD quality remain challenges [44].

Synthesis: Pedagogical strategies grounded in constructivism and supported by robust PD are foundational for content simplification [1] [3] [38].

4.2 Applications of Cognitive Load Theory

Worked Examples & Fading

- **Novice Learners:** Worked examples reduce cognitive load and anxiety, especially in mathematics and STEM [7] [33].
- **Fading:** Gradual removal of support enhances transfer and independent problem-solving [45] [46].
- **Process-Oriented vs. Product-Oriented:** Process-oriented examples foster deeper understanding and motivation in ill-structured domains [47] [48].

Multimedia Design Principles

- **Signaling & Segmenting:** Lower extraneous load and improve retention in digital learning [8] [49].
- **Redundancy & Split-Attention:** Instructional materials should avoid unnecessary repetition and integrate related information [6] [8].

Gamification & Task Segmentation

- **Motivation & Engagement:** Gamification increases engagement and can enhance learning outcomes, though effects on cognitive load are context-dependent [16] [17].
- **Segmenting Complex Tasks:** Improves performance in high-complexity domains when combined with adaptive feedback [18].

Synthesis: CLT-informed design is critical for simplifying content and optimizing learning, especially for novices and in digital environments [5] [7] [8].

4.3 Technology-Assisted Techniques

AI-Driven Adaptive Learning

- **Personalization:** AI tools like ChatGPT and intelligent tutoring systems adapt content and feedback to individual needs, supporting higher-order skills and inclusivity [9] [10] [11].
- **Generative AI:** Enables dynamic, context-aware content adaptation, though long-term impacts and ethical challenges require further study [12] [23].

Interactive Multimedia

- **Animations & Quizzes:** Enhance comprehension and engagement, especially in technical subjects [36] [37].
- **Accessibility Tools:** AI-powered and user-centered design approaches improve outcomes for students with disabilities [21] [22].

Ethical & Privacy Considerations

- **AI Chatbots:** Risks include bias, misinformation, and privacy violations; mitigation requires transparent design, bias detection, and adherence to data protection standards [50] [51].

Synthesis: Technology, especially AI, offers powerful tools for content simplification but must be implemented with attention to equity, ethics, and teacher readiness [9] [11] [12].

5. Real-World Implications

For Educators

- **Adopt Scaffolded, Inquiry-Based Methods:** Use learning progressions, case studies, and PBL to break down complex content [3] [40].
- **Leverage Worked Examples & Multimedia:** Integrate CLT principles in lesson and material design [7] [8].
- **Utilize Technology:** Incorporate AI-driven adaptive tools and interactive multimedia to personalize and simplify content [9] [36].
- **Prioritize Accessibility:** Ensure all materials meet accessibility standards and leverage assistive technologies [19] [22].

For Institutions

- **Invest in Teacher Professional Development:** Focus on inquiry-based, scaffolded, and technology-integrated PD [38].
- **Support Infrastructure for Technology Integration:** Provide resources and support for AI and multimedia adoption [52].
- **Develop Policies for Ethical AI Use:** Address privacy, bias, and transparency in AI-driven educational tools [50] [51].

For Students

- **Engage with Interactive Tools:** Use animations, quizzes, and AI tutors to supplement learning.
- **Self-Regulate Cognitive Load:** Apply strategies such as segmenting and self-explanation to manage complexity [53].

6. Future Research Directions

- **Longitudinal Impact of AI-Driven Simplification:** Examine sustained learning outcomes and knowledge retention across diverse populations [23] [24].

- **Optimization of Teacher PD:** Develop and evaluate PD models for effective integration of advanced technologies [38].
- **Comparative Effectiveness Studies:** Directly compare generative AI, traditional, and blended instructional approaches [12].
- **Inclusive Design for Neurodiverse & Multilingual Learners:** Expand research on content simplification strategies for these groups [22].
- **Ethical Frameworks for AI in Education:** Advance guidelines for privacy, bias mitigation, and equitable access [50] [51].

Conclusion and Recommendations

Summary of Effective Strategies

- **Constructivist Pedagogy:** Scaffolded, inquiry-based, and progression-aligned instruction is foundational for simplifying complex content [1] [3].
- **Cognitive Load Theory:** Worked examples, segmentation, and multimedia principles optimize comprehension and retention [5] [8].
- **Technology Integration:** AI-driven adaptive tools and interactive multimedia personalize and democratize access to simplified content [9] [36].
- **Accessibility & Inclusion:** Assistive technologies and user-centered design ensure all students benefit from content simplification [19] [22].
- **Teacher Professional Development:** Sustained, collaborative PD is essential for effective implementation [38].

Future Directions and Implementation

- **Scale Hybrid Models:** Combine teacher-led inquiry with real-time AI feedback for robust, scalable content simplification.
- **Institutional Support:** Prioritize investment in infrastructure, PD, and ethical frameworks for technology

adoption.

- **Continuous Evaluation:** Monitor and refine strategies based on student outcomes, teacher feedback, and emerging research.
- **Inclusive Innovation:** Ensure all content simplification efforts address the needs of diverse, multilingual, and neurodiverse learners.

In summary, breaking down complex educational content into accessible, "human language" formats is best achieved through a multi-dimensional strategy that integrates constructivist pedagogy, cognitive load theory, and technology-assisted personalization. Ongoing research, teacher development, and ethical vigilance are essential to maximize the benefits for all learners.

References

1. Transformative approaches: Exploring innovative pedagogies for 6G education Ochoa-Tataje, F., Alanya-Beltran, J., Ruiz-Salazar, J., (...), Perez-Mendoza, A. 6G Security Education and Multidisciplinary Implementation, 2024
https://www.scopus.com/pages/publications/85202591653?origin=scopusAI

2. A Liberating Education: Integrating Funds of Knowledge and Disciplinary Knowledge to Create Tools for Students' Lives Castillo-Montoya, M., Ives, J. New Directions for Teaching and Learning, 2020
https://www.scopus.com/pages/publications/85099520354?origin=scopusAI

3. Teaching First Graders to Comprehend Complex Texts Through Read-Alouds Witte, P.G. Reading Teacher, 2016
https://www.scopus.com/pages/publications/84966663972?origin=scopusAI

4. When Five Steps Changed Everything: An Action Research on SQ3R Implementation for Reading Comprehension Mastery Dwiniasih, D., Aisyah, A., Purnama, A.P.A.G., (...), Suwarno, S. Reading Psychology, 2026
https://www.scopus.com/pages/publications/105043711133?origin=scopusAI

5. Optimising student cognitive load in computer education Tuovinen, J.E. ACM International Conference Proceeding Series, 2000
https://www.scopus.com/pages/publications/85009129279?origin=scopusAI

6. Levels of expertise and instructional design Kalyuga, S., Chandler, P., Sweller, J. Human Factors, 1998

https://www.scopus.com/pages/publications/0032032804?origin=scopusAI

7. Worked Examples Moderate the Effect of Math Learning Anxiety on Children's Math Learning and Engagement During the COVID-19 Pandemic Mesghina, A., Vollman, E., Trezise, K., Richland, L.E. Journal of Educational Psychology, 2024 https://www.scopus.com/pages/publications/85168834788?origin=scopusAI

8. Principles for Reducing Extraneous Processing in Multimedia Learning: Coherence, Signaling, Redundancy, Spatial Contiguity, and Temporal Contiguity Principles Fiorella, L., Mayer, R.E. The Cambridge Handbook of Multimedia Learning, 2021 https://www.scopus.com/pages/publications/105028065476?origin=scopusAI

9. AI-Supported Learning: Integrating ChatGPT to Enhance Cognitive Skills in STEM Education Sánchez-Rodriguez, E., Merino-Soto, C., Chen, M.H., (...), Chans, G.M. IEEE Global Engineering Education Conference, EDUCON, 2025 https://www.scopus.com/pages/publications/105008193918?origin=scopusAI

10. An introduction to generative AI tools for education 2030 Sandhu, R., Channi, H.K., Ghai, D., (...), Kaur, M. Integrating Generative AI in Education to Achieve Sustainable Development Goals, 2024 https://www.scopus.com/pages/publications/85198175042?origin=scopusAI

11. Using an intelligent tutoring system as a textbook supplement for personalized learning McIntosh, D., Ferguson, S.A.K. International Journal of Technologies in Learning, 2017 https://www.scopus.com/pages/publications/85040118018?origin=scopusAI

12. Pedagogical transformation through generative AI: a hybrid deep learning comparison with traditional digital learning materials based on GenAI-EduNet framework Alshammari, F. Quality and Quantity, 2026 https://www.scopus.com/pages/publications/105034414866?origin=scopusAI

13. Developing nurse educator competency in the pedagogy of simulation McNeill, J., Parker, R.A., Nadeau, J., (...), Cook, J. Journal of Nursing Education, 2012 https://www.scopus.com/pages/publications/84870656825?origin=scopusAI

14. Multi-faceted professional development designs for science teachers' self-efficacy for inquiry-based teaching: A critical review Seneviratne, K., Hamid, J.A., Khatibi, A., (...), Sudasinghe, S. Universal Journal of Educational Research, 2019 https://www.scopus.com/pages/publications/85071190273?origin=scopusAI

15. From training to practice: the lasting effects of IBSE on physics teachers' professional development and classroom practices Capone, R. Physics Education, 2025 https://www.scopus.com/pages/publications/105010911291?origin=scopusAI

16. The effect of AI-enhanced gamification on learning outcomes in higher education Kaur, R., Sarkar, R., Lalitha, M., (...), Anand, T. Digital Transformation and Sustainability in Higher Education, 2025 https://www.scopus.com/pages/publications/105012674565?origin=scopusAI

17. Gamification Effect: Enhancing Student Learning Outcomes in Higher Education: A Meta-Analysis and Policy Implications Gyedu, F.O., Partey, P.A., Boafo, F.A., Agbo, D.D. SAGE Open, 2026 https://www.scopus.com/pages/publications/105030685882?origin=scopusAI

18. Using gamification to teach forecasting in a business school setting Legaki, N.Z., Karpouzis, K., Assimakopoulos, V. CEUR Workshop Proceedings, 2019 https://www.scopus.com/pages/publications/85065698898?origin=scopusAI

19. Technology Tools to Promote Digital Accessibility Caprette, H. Guide to Digital Accessibility: Policies, Practices, and Professional Development, 2023 https://www.scopus.com/pages/publications/85167744589?origin=scopusAI

20. Advancing Inclusive Education and Disability Support Through Technology and Innovation Jacob, U.S., Maguvhe, M.O. Advancing Inclusive Education and Disability Support Through Technology and Innovation, 2026 https://www.scopus.com/pages/publications/105046216165?origin=scopusAI

21. A Multimodal Approach to Enhance Cognitive Accessibility in Educational Technologies Krishnaraj, M., Serrena, J., Shreen Thasbeeha, O.M.R. CCIC 2026 - Contemporary Computing Innovations Conference 2026, 2026 https://www.scopus.com/pages/publications/105038760165?origin=scopusAI

22. Research Trends in Information Accessibility to Web Content for Adults with Developmental Disabilities: A Literature Review Seino, K. Lecture Notes in Computer Science, 2025 https://www.scopus.com/pages/publications/105002010701?origin=scopusAI

23. The Impact of Generative Artificial Intelligence on Student Learning Outcomes: A Second-Order Metaanalysis Zhuang, X., Liu, B., Sun, Y., (...), Liu, B. 2026 8th International Conference on Computer Science and Technologies in Education, CSTE 2026, 2026 https://www.scopus.com/pages/publications/105048935525?origin=scopusAI

24. The Potential of Generative AI in Supporting Neurodiversity in Higher Education: A Systematic Review Liao, J., Lee, C.S. Proceedings of the Association for Information Science and Technology, 2025 https://www.scopus.com/pages/publications/105019384551?origin=scopusAI

25. Staying connected: online education engagement and retention using educational technology tools. Salazar, J. Clinical laboratory science : journal of the American Society for Medical Technology, 2010 https://www.scopus.com/pages/publications/77957273273?origin=scopusAI

26. STEM-5E Socio-Scientific Argumentation with Generative AI-Driven Scaffolding: Exploring the Interplay between Epistemic Beliefs and Learning Outcomes Sui, C.-J., Chang, C.-Y., Yen, M.-H. Journal of Science Education and Technology, 2026 https://www.scopus.com/pages/publications/105017818061?origin=scopusAI

27. Reinventing STEM null Reinventing STEM in Early Childhood Education: A Constructivist Approach, 2025 https://www.scopus.com/pages/publications/105003990413?origin=scopusAI

28. An Exploratory Teaching Proposal of Greek History Independence Events based on STEAM Epistemology, Educational Robotics and Smart Learning Technologies Kourtis, S., Xenakis, A., Kalovrektis, K., (...), Chalvantzi, I. ACM International Conference Proceeding Series, 2021 https://www.scopus.com/pages/publications/85127458893?origin=scopusAI

29. Leveraging Learning Progressions to Enhance Pedagogical Content Knowledge in Science Education Ramaila, S. Current Trends and Best Practices of Pedagogical Content Knowledge (PCK), 2025 https://www.scopus.com/pages/publications/105007381059?origin=scopusAI

30. Empirical recovery of learning progressions through the lens of educators Ketterlin-Geller, L.R., Zannou, Y., Sparks, A., Perry, L. Journal of Mathematical Behavior, 2020 https://www.scopus.com/pages/publications/85090601660?origin=scopusAI

31. A case study on genomic imprinting facilitates student learning Goudsouzian, L.K., Lo, S.M. Advances in Physiology Education, 2023 https://www.scopus.com/pages/publications/85147720521?origin=scopusAI

32. Investigating the Effectiveness of Problem Based Learning on Academic Achievement in EFL Classroom: A Meta-Analysis Orhan, A. Asia-Pacific Education Researcher, 2025 https://www.scopus.com/pages/publications/85197537453?origin=scopusAI

33. Suggestions for graduated exposure to programming concepts using fading worked examples Gray, S., St. Clair, C., James, R., Mead, J. Third International Computing Education Research Workshop, ICER'07, 2007
https://www.scopus.com/pages/publications/38849186591?origin=scopusAI

34. Effects of segmenting, signalling, and weeding on learning from educational video Ibrahim, M., Antonenko, P.D., Greenwood, C.M., Wheeler, D. Learning, Media and Technology, 2012
https://www.scopus.com/pages/publications/84866872991?origin=scopusAI

35. Evaluating the Effects of Educational Multimedia Design Principles on Cognitive Load Using EEG Signal Analysis Farkish, A., Bosaghzadeh, A., Amiri, S.H., Ebrahimpour, R. Education and Information Technologies, 2023
https://www.scopus.com/pages/publications/85137476354?origin=scopusAI

36. Empowering STEM educators with digital tools Papadakis, S., Kalogiannakis, M. Empowering STEM Educators With Digital Tools, 2024
https://www.scopus.com/pages/publications/105008604727?origin=scopusAI

37. Java applets to reinforce fundamental computer science concepts Quinn, M.J. Computers in Education Journal, 2006
https://www.scopus.com/pages/publications/33749548262?origin=scopusAI

38. Pedagogical Strategies for the Development of Inquiry-Based Learning in Teacher Education: A Systematic Review of the Literature from 2014 to 2024 Saavedra Carrasco, J.M., Ardiles Hidalgo, I.L., Saavedra Carrasco, L.A., (...), del Rosario Vasquez Barboza, E. International Journal of Learning in Higher Education, 2025
https://www.scopus.com/pages/publications/105008237150?origin=scopusAI

39. Federated Learning for Accessible Engineering Education: A Survey on AI-Driven Personalization for Students with Disabilities Mohammed, B., Noureddine, S. International Journal of Engineering Education, 2025
https://www.scopus.com/pages/publications/105021880194?origin=scopusAI

40. Developing a Cohesive Active Learning Approach by Integrating Theoretical Case Studies and Practical Problem-Based Learning Principles Otieno, F.X. Translational Systems Sciences, 2019
https://www.scopus.com/pages/publications/105034676758?origin=scopusAI

41. A Bayesian Network Meta-Analysis to Synthesize the Influence of Contexts of Scaffolding Use on Cognitive Outcomes in STEM Education Belland, B.R., Walker, A.E., Kim, N.J. Review of Educational Research, 2017

https://www.scopus.com/pages/publications/85032927817?origin=scopusAI

42. A systematic review of research on project-based learning in reducing misconceptions in Science Learning Haryono, H.E., Jatmiko, B., Prahani, B.K., Wahab, M.N.N.D. Journal of Curriculum Studies Research, 2026 https://www.scopus.com/pages/publications/105052250140?origin=scopusAI

43. Inquiry-based science education: Tools for supporting the design of schoolteacher professional development programs Akuma, F.V., Koenen, J. Eurasia Journal of Mathematics, Science and Technology Education, 2025 https://www.scopus.com/pages/publications/85219583344?origin=scopusAI

44. Behind the scenes: teachers' perspectives on factors affecting the implementation of inquiry-based science instruction Baroudi, S., Rodjan Helder, M. Research in Science and Technological Education, 2021 https://www.scopus.com/pages/publications/85070511369?origin=scopusAI

45. The consequences of fading instructional guidance on delayed performance: The case of financial services training Kissane, M., Kalyuga, S., Chandler, P., Sweller, J. Educational Psychology, 2008 https://www.scopus.com/pages/publications/55849123994?origin=scopusAI

46. Toward a fundamental understanding of worked example instruction: Impact of means-ends practice, backward/forward fading, and adaptivity Moreno, R., Reisslein, M., Delgoda, G.M. Proceedings - Frontiers in Education Conference, FIE, 2006 https://www.scopus.com/pages/publications/48749083304?origin=scopusAI

47. Exploring the impact of concept-oriented faded WOE and metacognitive scaffolding on learners' transfer performance and motivation in programming education Shin, Y., Jung, J., Lee, H.J. Metacognition and Learning, 2024 https://www.scopus.com/pages/publications/85175566138?origin=scopusAI

48. Process-oriented worked examples: Improving transfer performance through enhanced understanding Van Gog, T., Paas, F., Van Merriënboer, J.J.G. Instructional Science, 2004 https://www.scopus.com/pages/publications/3843109054?origin=scopusAI

49. Graphic organizers' optimized design through segmenting and signaling principle: Based on generative learning theory Qin, T., Wu, C.-J. Computers and Education, 2025 https://www.scopus.com/pages/publications/105012622495?origin=scopusAI

50. Talking to Strangers: The Challenges of Conversational AI and Human Expectations Lachman, R. Revolutionizing

Communication: the Role of Artificial Intelligence, 2024 https://www.scopus.com/pages/publications/85209848271?origin=scopusAI

51. Standardizing and Scaffolding Health Care AI-Chatbot Evaluation: Systematic Review Hua, Y., Xia, W., Bates, D., (...), Torous, J. JMIR AI, 2025 https://www.scopus.com/pages/publications/105023084901?origin=scopusAI

52. The Transformative Influence of Generative AI on Teaching and Learning Mavundla, K., Abayomi, A., Adetiba, E., (...), Thakur, S. Lecture Notes on Data Engineering and Communications Technologies, 2025 https://www.scopus.com/pages/publications/105011959834?origin=scopusAI

53. The Cognitive Load Self-Management Principle in Multimedia Learning Zhang, S., de Koning, B., Agostinho, S., (...), Paas, F. The Cambridge Handbook of Multimedia Learning, 2021 https://www.scopus.com/pages/publications/105028046702?origin=scopusAI